

Semantic Segmentation-Based Occupancy Grid Map Learning With Automotive Radar Raw Data

Yi Jin , *Graduate Student Member, IEEE*, Marcel Hoffmann , *Graduate Student Member, IEEE*, Anastasios Deligiannis , *Member, IEEE*, Juan-Carlos Fuentes-Michel , and Martin Vossiek , *Fellow, IEEE*

Abstract—Precise road scene understanding is of great essence to autonomous driving. As a widely used method for road scene understanding, occupancy grid mapping is leveraged to detect obstacles and predict drivable road areas. Because of its robustness under harsh conditions, low cost, and large perceptual range, radar sensor is becoming increasingly important to achieve various critical perception tasks. However, for radar-based occupancy grid mapping, current inverse sensor model (ISM) relies on detection data and is most hand-crafted. In this work, we propose a novel data-driven ISM that employs the range-Doppler matrix as the input. With a systematic evaluation and comparison of our model with classic, hand-crafted ISM and the data-driven, detection-based Occupancy Net using RADIAL dataset, we find that data-driven models are far superior to their hand-crafted counterpart. Furthermore, although both data-driven models are on par within near range (< 50 m), our model outperforms Occupancy Net by a large margin in far range ($[50$ m, 100 m]). Specifically, our model has about a 0.3 and 0.4 improvement in distant range for the intersection over union (IoU) and F_1 score, respectively. In addition, leveraging adjacent occupancy grid map prediction, we propose a radar-based occupancy flow to precisely distinguish moving objects.

Index Terms—Automotive radar, occupancy grid map, inverse sensor model, road segmentation, deep learning, transformer.

I. INTRODUCTION

CURRENTLY, autonomous driving (AD) is considered one of the most promising technologies, especially with the booming of artificial intelligence, or more specifically, deep learning [1]. To realize AD, autonomous vehicles (AVs) must be able to understand their surroundings like a human being, and their perception system is responsible for this task. A massive deployment of AVs in daily life faces two main challenges. First, the driving conditions for a vehicle are usually complicated and various, and the perception system must work reliably, even in inclement conditions, such as fog, snow, and dark [2]. Second, because of the inherent, long-tail issue of deep learning

algorithms, a road user classification could fail if the case is not present or rare in the training data [3], which poses a large hazard to human life and property.

Regarding the first challenge, compared with other sensors such as camera and lidar, a radar sensor is most reliable because it is capable of working robustly under adverse conditions and harsh environments [4]. In this regard, the radar sensor is deemed a key component in the perception system. Furthermore, although the enlargement of a dataset or collecting specific data targeting the failure case is a common practice to alleviate the long-tail issue, these approaches cannot eliminate it totally. In addition, since a long-tail case happens infrequently, one observation will likely require hundreds of millions of miles measurement, which is obviously time-consuming and resource-intensive [3]. Elluswamy et al. [5] contended that the geometry is more important than the ontology, and they designed the vision-based Occupancy Network to predict the drivable area that can assist the vehicle to avoid a collision with unrecognized objects. In light of these results, this work focuses on occupancy grid learning with radar data.

Occupancy grid mapping [6] refers to the prediction of obstacles on the road, including not only static but also dynamic objects, which is regarded as the basic part of understanding the road scene. Radar-based occupancy grid mapping has been a crucial research area for a while. Most pertinent works are based on the classic method illustrated in Fig. 1(b), such as [7], [8], [9], [10]. Since radar detection of one frame is extremely sparse, Werber et al. [8] accumulated the detection from multiple frames and harnessed a hand-crafted ISM designed by human experts who rely on their understanding of the sensor characteristics and the observed environment, to yield the occupancy grid map (OGM). However, this method usually cannot correctly interpret the radar data when the noise or clutter is strong or under complex situations. Furthermore, with the advance of radar technology, the assumption that each cell in the OGM is independent is no longer satisfied because the detections from an object usually span multiple neighboring cells in the radar measurement. Therefore, Sless et al. proposed a data-driven ISM called Occupancy Net that learns the model from the given data, using this to replace the hand-crafted counterpart and yielded better road segmentation performance [11]. Nowruzi et al. tested the performances of data-driven ISM with the input of range-Doppler-azimuth tensor or range-azimuth map, but their measurements and results were only restricted to parking scenarios [12]. In addition, works such as [13] and [14], proved

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Yi Jin, Marcel Hoffmann, and Martin Vossiek are with the Institute of Microwaves and Photonics, Friedrich-Alexander-Universität Erlangen-Nürnberg, 91058 Erlangen, Bavaria, Germany (e-mail: yi.y.jin@fau.de; marcel.mh.hoffmann@fau.de; martin.vossiek@fau.de).

Anastasios Deligiannis and Juan-Carlos Fuentes-Michel are with BMW Group, 80788 Munich, Germany (e-mail: anastasios.deligiannis@bmw.de; juan-carlos.fuentes-michel@bmw.de).

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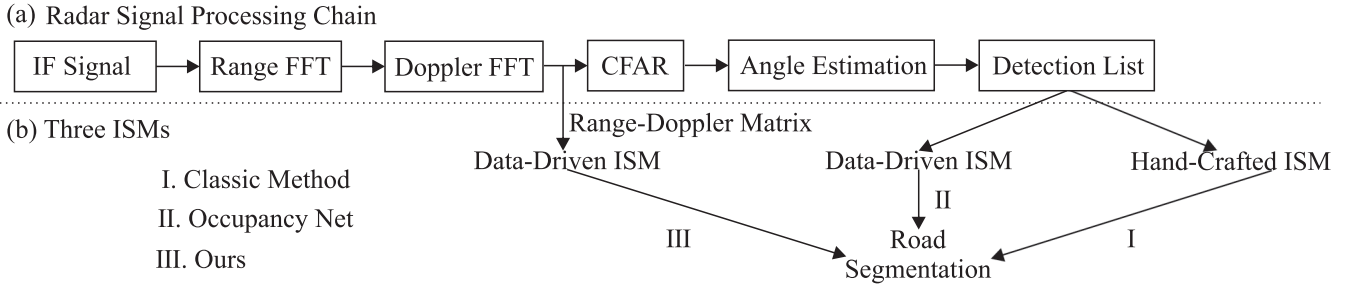


Fig. 1. Subfigure (a) shows radar signal processing chain. Subfigure (b) illustrates three ISMs to achieve road segmentation, which requires recognizing not only the static but also dynamic obstacles. I and II leverage detection as the input that is generated by constant false alarm rate (CFAR) and angle estimation from a range-Doppler matrix, while our method employs range-Doppler matrix directly.

the superiority of the data-driven ISM. Kung et al. even utilized the penetrating capability of radar to interpret the scene behind the obstacle [14]. Although Orr et al. designed a network with radar raw data to perceive the road in [15], their results and evaluation were in the image view instead of the favorable bird's eye view (BEV). The discrepancy between both views makes the comparison results less rigorous, which can be confirmed by our findings. Moreover, the image view suffers from the inherent scale problem, and the transformation to BEV is ill-posed [16], which is not convenient for the direct application or the following downstream tasks, such as trajectory estimation.

Considering these issues, in this work, we proposed a novel data-driven ISM that is based on radar raw data to achieve precise road segmentation in BEV. Furthermore, we utilized a cross-modal supervision method employing camera and lidar data to generate accurate BEV road annotation for RADIAL dataset [17] and extensively evaluated and compared three ISMs. With this work, we endeavoured to answer the questions regarding whether it is necessary to utilize raw data-based ISM and, if so, when.

Our contributions in this work can be summarized in the following three aspects:

- We proposed a novel radar raw data-based, data-driven ISM to achieve precise road segmentation, including static and dynamic obstacles, in BEV. Furthermore, the proposed network with a transformer-based encoder, transcends all previous works obviously.
- To the best of our knowledge, we are the first to conduct comprehensive evaluation and comparison of all three ISMs. Based on the results, we confirmed that data-driven ISMs are far superior to their hand-crafted competitor. Furthermore, between two data-driven ISMs, we discovered that the raw data-based approach outperforms the detection-based counterpart significantly in the far range (> 50 m), while in the near range, both are on par.
- Employing multiple adjacent frames, we designed a novel radar-based occupancy flow algorithm that can recognize moving objects from OGM.

This work is organized as follows: Section II introduces the occupancy grid mapping, radar signal processing, the employed RADIAL dataset, and the transformer structure; Section III describes how we prepared and processed the ground truth data; from Sections IV and V, we first explained the proposed novel

ISM network in detail, then comprehensively compared three types of ISMs depicted in Fig. 1(b), and analyzed our model with other data-driven models systematically; in Section VI, we elaborated on the radar-based occupancy flow algorithm and summarized the comparison of three ISMs; finally, we concluded the work in Section VII.

II. RELATED WORKS

A. Radar Signal Processing

Due to its low cost, precise range and velocity perceptual capability, and robustness under inclement conditions, automotive radar is regarded as a key component for advanced driver assistance systems (ADAS) as well as the coming AD [2]. To satisfy the increasing requirement on resolutions under the limited hardware and size conditions, the multiple-input and multiple-output (MIMO) technology is widely adopted in current frequency-modulated continuous wave (FMCW) automotive radars, and usually, the required orthogonality is achieved by methods such as time-division multiplexing (TDM), Doppler-division multiplexing (DDM) and frequency-division multiplexing (FDM) [18].

The state-of-the-art signal processing chain of the FMCW automotive radar is depicted in Fig. 1(a). First, the window function and fast Fourier transform (FFT) operation are conducted along the fast time direction of intermediate frequency (IF) signal, and this step is called Range FFT. Then, similar operations, named Doppler FFT, are performed along the slow time, so that the obtained result corresponds to a range-Doppler matrix. Afterwards, a CFAR [19] is employed to detect peaks in the range-Doppler matrix. More specifically, the signal power of cell under test (CUT) is compared with a threshold value, which is derived from the corresponding reference cells. If its signal is larger than the threshold, a peak is detected. Afterwards, the azimuth angle of detection along with the elevation for a 4D radar is estimated. Digital beamforming [20], [21], which is based on FFT, is an efficient method to estimate the angle. However, the angular resolution of digital beamforming is rather limited, compared with that of other high-resolution methods such as multiple signal classifier (MUSIC) [22], estimation of signal parameters via rational invariance technology (ESPRIT) [23], sparse sensing-based methods [24], [25] and iterative adaptive approach (IAA) [26]. The subspace-based MUSIC algorithm

requires an accurate estimation of the array covariance matrix with multiple snapshots, which can be challenging for highly dynamic automotive scenarios. In this regard, spatial smoothing [27] can be employed to introduce virtual snapshots to solve this issue. In contrast, sparse sensing-based methods and IAA achieve angle estimation with a single snapshot, but it incurs a high computational load. After the angle estimation, a detection list is generated.

Compared with the output of a lidar sensor, radar detections of one cycle are still rather sparse. The sparse single snapshot data are detrimental to a precise, holistic perception, especially in the far range, which can be confirmed by this work.

B. Occupancy Grid Map

As a representation of the environment, OGM was first proposed by the robotics community several decades ago [6], [28]. Nowadays, it is frequently employed in AD area [8], [29], [30], because of its high efficacy.

To produce the occupancy grid map, the space around the vehicle is discretized into a range of cells with spatial resolution α_x and α_y . Each cell has a corresponding random variable to represent its occupancy status. Assuming a grid map $\mathbf{m}$ with size $M \times N$, $\mathbf{m}_{u,v}$ denotes the value of the occupancy grid map $\mathbf{m}$ at position (u, v) . If $\mathbf{m}_{u,v} = 1$, this means *occupied*, while $\mathbf{m}_{u,v} = 0$ represents *free*. Mathematically, the aim of the occupancy grid map algorithm is to estimate the posterior likelihood

$$p(\mathbf{m} = 1 | z_{1:t}, x_{1:t}) \quad (1)$$

where $z_{1:t}$ and $x_{1:t}$ represent the sensor measurements and vehicle poses, respectively, from the moment 1 to t . In addition, (1) can be simplified as $p(\mathbf{m} | z_{1:t}, x_{1:t})$.

Conventionally, the occupancy grid map has two cornerstones, that is the ISM and Bayesian filtering [31]. A ISM predicts the *occupancy* likelihood of the cells based on a single measurement, and the Bayesian filtering processes the prediction from ISM recursively. Direct estimation of (1) is computationally intractable, therefore the assumption that the state of each cell is independent with each other is proposed to simplify the issue. As a result, (1) can be decomposed into the following:

$$p(\mathbf{m} | z_{1:t}, x_{1:t}) = \prod_{u,v} p(\mathbf{m}_{u,v} | z_{1:t}, x_{1:t}) \quad (2)$$

Since too small or too large likelihood could cause numerical instability during calculation, a log-odd representation is preferred:

$$l_t(\mathbf{m}_{u,v} | z_{1:t}, x_{1:t}) = \log \frac{p(\mathbf{m}_{u,v} | z_{1:t}, x_{1:t})}{1 - p(\mathbf{m}_{u,v} | z_{1:t}, x_{1:t})} \quad (3)$$

Furthermore, when applying the Bayesian filtering to update the occupancy likelihood at cell (u, v) recursively, the following is derived:

$$\begin{aligned} l_t(\mathbf{m}_{u,v} | z_{1:t}, x_{1:t}) &= l_t(\mathbf{m}_{u,v} | z_t, x_t) \\ &+ l_{t-1}(\mathbf{m}_{u,v} | z_{1:t-1}, x_{1:t-1}) - l_0 \end{aligned} \quad (4)$$

where $l_0 = \log \frac{p(\mathbf{m}_{u,v}=1)}{p(\mathbf{m}_{u,v}=0)}$ as the occupancy log-odd prior is usually zero, and the term $l_t(\mathbf{m}_{u,v} | z_t, x_t)$ denotes the ISM.

When it comes to the automotive radar-based OGM, most available works are based on hand-crafted ISM and employ radar detection as the input. For instance, Schmid et al. [32] proposed a hierarchical, 3D occupancy grids method utilizing static detection to predict the parking space geometry. Werber et al. [8] proposed utilizing a hand-crafted ISM, which considered the radar-tailored detection plausibility in range, azimuth and signal intensity, to predict the occupancy state.

Generally, there are three issues with this method. First, the hand-crafted ISM sometimes cannot correctly interpret the radar measurement because radar data are inherently rather noisy [33], and its detection is considerably sparse, compared with other sensors such as lidar. To mitigate the problem caused by sparse detection, Prophet et al. [9] implemented an adaption to the classic occupancy grid map algorithm that determines the extra free space between adjacent detections. Li et al. [34] utilized clustering algorithms to filter the noise and clutter. Slutsky et al. [35] extended the common ISM to positive and negative components, and the negative component mitigated noise signals. However, these detection-based methods only alleviated the issue but cannot fix it thoroughly. As the works [36], [37], [38], [39] show impressive results with radar raw data, which includes range-Doppler and range-azimuth matrices and even 3D radar cube, for object detection and segmentation, we opt for the range-Doppler matrix instead of the detection as the input in this work. Considering that the radar raw data contain much more copious information than detections, these data are promising to eliminate the issues caused by inherent noise and sparse detection. In this context, Nowruzi et al. [12] investigated the influence of different radar raw data on free space segmentation task, which is similar to occupancy grid mapping, but their results were limited to parking scenarios.

Second, usually, only static detection can be processed by this approach, and the crucial moving detection has to be abandoned. Since radar detections from different times must be accumulated to generate the occupancy grid map considering the motion of the ego vehicle, the accumulation of the moving detection causes discrepancies. Nuss et al. used a random finite set approach to handle moving objects [40]. However, this approach was computationally intensive and dedicated to dynamic detection, which means that two different frameworks must be implemented to deal with static and dynamic detections respectively. Last but not least, the detection of an object, such as a vehicle, by current automotive-grade radars spans multiple neighboring cells; thus, the independent assumption for the cells in OGM cannot be satisfied anymore. In this regard, Sless et al. [11] proposed a data-driven ISM to learn the occupancy grid mapping, and consequently achieved better results than the hand-crafted counterpart. In addition, not only a sophisticated ISM but also a heteroscedastic uncertainty considering radar sensor features and interactions with the environment are learned by the model in [13]. Kung et al. [14] proposed a Unet-based network to predict the probabilistic distribution of occupancy status in a polar coordinate system and their model even preserved the

TABLE I
RADAR DATASET COMPARISONS

Dataset	Num. of Frames	Radar	Radar Data Type	Other Sensors	Diversity
NuScenes [42]	40,000	5 Surrounding Radars	2D PC	Camera & Lidar	U
Oxford Radar [43]	240,000	1 Rotating Radar (No Doppler)	RA	Camera & Lidar	V & U
RadarScenes [44]	1,556,684	4 Surrounding Radars	2D PC	Camera	U
Astyx [45]	500	1 Front Radar	3D PC	Camera & Lidar	U
View-of-Delft [46]	8,693	1 Front Radar	3D PC	Camera & Lidar	U
CARRADA	12,666	1 Front Radar	RAD	Camera	T
RADial [17]	8,252	1 Front Radar	3D PC & ADC	Camera & Lidar	U, R & H
K-Radar [47]	35,000	1 Front Radar	4D RT	Camera & Lidar	V

1. In the Radar Data Type column, PC stands for point cloud, RA for range-azimuth matrix, RAD for range-azimuth-Doppler tensor, ADC for analog-digital-converter signal, and RT for radar tensor.

2. In the Diversity column, U stands for urban scenes, V for various lighting and weather situations, T for test tracks, R for rural scenes, and H for highway scenarios.

penetration capability of radar. Compared with the hand-crafted counterpart, a data-driven ISM is more robust and can interpret the surroundings more precisely when handling noisy and sparse detection. Therefore, in this work, we propose our data-driven model to address the occupancy grid mapping issue. Considering that previous works [11], [13], [14] do not handle moving objects properly due to a static environment assumption, we propose a radar-based occupancy flow algorithm to address it.

Apart from an occupancy status estimation, a higher level environment understanding, such as semantic level, can be achieved with OGM. For instance, Schumann et al. [10] employed a convolutional neural network to segment the static surroundings. Prophet et al. [41] utilized simulated and real measurements to compare the segmentation results based on 2D and 3D OGM. Although a finer semantic environment understanding including vehicles, infrastructure, etc. is out of the scope of this work, it can be a promising direction for future work.

C. RADial Dataset

A brief comparison of several current radar datasets is listed in Table I. Apart from datasets like NuScenes [42] and RadarScenes [44] containing only radar detection, several datasets provide low-level radar data. For instance, CARRADA [48] includes 3D radar cubes, but the measurements were limited to simple test tracks and the radar was stationary. In contrast, K-Radar [47] provides 4D radar tensors with synchronized lidar and camera data in various lighting and weather situations on the street. However, as the influence of challenging conditions is not the focus of this work, we finally selected the RADial dataset [17].

RADial dataset is a newly published dataset that includes synchronized camera, radar and lidar data. Concerning the radar data, not only the detection but also the analog-to-digital converter (ADC) signal are provided, which largely facilitates the research of the entire community. Three different scenarios, namely urban road, country road, and highway, are covered in the dataset, and the entire measurement that is made up by 91 sequences of one to four minutes duration lasts two hours. In this work, we opted for the ready-to-use part with 8252 frames from the total 25000 synchronized frames, and most road users in the selected part are vehicles.

The radar of this dataset is a prototype high-definition (HD) front radar based on the Doppler-division multiplexing with 12 transmitter antennas and 16 receiver antennas. The radar

is alleged to have 0.2 m range resolution, 0.1° and 1° angle resolutions in azimuth and elevation directions, respectively. Apart from the radar, a 5 Mpix resolution camera and an automotive-grade lidar with 16 layers are mounted on the vehicle to have a common field of view (FoV) with the radar. For more details about the dataset, readers can refer to [17].

D. Transformer

Bahdanau et al. first proposed attention mechanism [49], which is designed to learn an alignment between source and target sequences for machine translation. Afterwards, Vaswani et al. [50] introduced the first attention-based encoder-decoder structure—transformer in the natural language processing area, which has an excellent capability to acquire long-range dependency. The application of a transformer in the vision area, called Vision Transformer (ViT), is first proposed by Dosovitskiy et al. [51] for image classification. Despite the strengths of ViT against other structures such as convolutional neural network (CNN), the immense computational load required by the global attention in [51] largely constricts its application to other tasks, such as segmentation or detection. Swin Transformer [52], which applied self-attention within each small window of the entire image and hierarchical feature maps, solved this issue and utilized a shift window to build up cross window information exchange. In this work, we follow the design ethos of the Swin Transformer, which is based on window-based multihead self-attention (W-MSA), and simultaneously consider the features of the range-Doppler matrix and the adopted DDM to design the preencoder in the network.

III. DATA PREPARATION

To prepare the OGM annotation in the BEV, authors of [11] and [13] turned to the lidar data due to its distinguished 3D perception capability. Considering that a single lidar frame projected to the grid map is still sparse, especially for a far range, the generated ground truth cannot reflect the holistic surroundings and consequently degrades the following learning process. In this regard, Sless et al. proposed to aggregate lidar point cloud from adjacent frames to address it effectively [11]. However, RADial dataset does not provide the complete odometry information necessary for the aggregation procedure. Therefore, we combined the camera image and lidar data to generate the ground truth $\mathcal{G}$, and Fig. 2(c) shows one example of $\mathcal{G}$. Herein, $\mathcal{G} \in \{0, 1\}^{H \times W}$ with $H = 512$ and $W = 256$, is defined in the

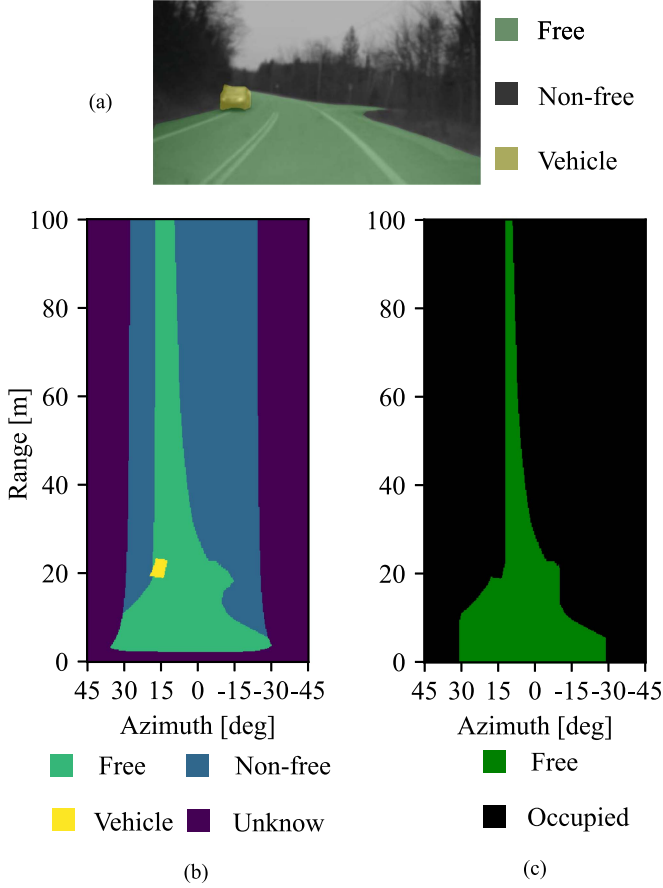


Fig. 2. Example illustrates the process of OGM annotation generation. (a) shows a panoptic segmentation example. (b) is the projection result from the segmented image (a) and the bounding box of vehicle from lidar data in the polar coordinate system. (c) is the corresponding OGM annotation converted from (b).

polar coordinate system of ego-vehicle covering the area of range $R_{\text{ogm}} \in [0, 100 \text{ m}]$ and azimuth $\alpha_{\text{ogm}} \in [-45^\circ, 45^\circ]$, where a value of 1 represents the *free* class that is not only semantically free but also satisfies the line of sight (LOS) condition, and 0 denotes the *occupied* class, including all the other pixels. Instead of the usual representation of OGM in the Cartesian coordinate system, we adopted the polar coordinate system for the following three reasons. First, a polar representation should be inherently closer to the radar data since they are originally in a polar coordinate system. In addition, the processing, such as the determination of the free space from the origin to obstacles and occlusion problem, is much easier and more efficient in this coordinate system. Last but not least, although the occupied and unknown statuses are not explicitly distinguished, these two statuses can be inferred in the polar representation. More specifically, assuming that we do not consider the penetration capability of the radar, for a given azimuth angle, the first cell assigned the *occupied* class that changes from the *free* is actually occupied by obstacles, and the following cells are unknown. Consequently, occupied and unknown statuses are reconciled in the polar coordinate system. Since precise annotations behind obstacles are rather challenging with camera and lidar

data, our practice should be acceptable. However, the polar coordinate system has the flaw that it is arduous to exploit temporal information from predicted OGM sequences. Thus, an extra conversion to the Cartesian is necessary, and this is elaborated in Section VI-A.

The entire process consists of three steps. First, we projected panoptically segmented camera images, which contain free space, non-free space, and vehicle instances classes, to the BEV grid maps. Nevertheless, in this step, the positional deviation of vehicles was rather obvious considering their size. In the second step, we converted bounding boxes of vehicles to the grid map to refine their positional accuracy. To obtain the bounding boxes, we employed CenterPoint [53] to segment the corresponding lidar point cloud, and a vehicle prediction was confirmed only when the predictions from the camera and lidar data were consistent with each other. At last, the semantically segmented grid map was adjusted to the OGM annotation. As an illustration, subfigures of Fig. 2 show the segmented camera image, the projection in the polar BEV, and the finally generated OGM annotation, alphabetically.

Usually, a vision-based annotation method is relatively precise only in the near range. However, we took several methods to ensure a good annotation quality in the entire scope $[0 \text{ m}, 100 \text{ m}]$. To guarantee the quality of image segmentation, we manually corrected the prediction error of the segmented images generated by panoptic FPN [54], and put extra effort into the modification of the road boundary area. These effects can be observed in Fig. 2(a). Considering that the projection of vehicles from camera images to BEV incurs relatively large positional errors compared with their size, their bounding boxes provided by lidar data are leveraged to determine the exact positions. Moreover, when adjusting to our customized polar BEV annotation, namely from Fig. 2(b) and (c), since areas behind the nearest obstacle are considered occupied, the annotation requirement for distant range can be alleviated. Apart from these methods, factors that, in RADIAL dataset, the road shape and traffic situation are not rather complicated within a 100 m range, also facilitate our cross-modal supervision method to keep high precision in the distant range.

IV. PROPOSED METHOD

In this work, the proposed network consists of an encoder and a decoder, presented in Fig. 3(a). The encoder is a self-attention-based transformer structure and made up of a DDM Preencoder as well as a Swin Transformer backbone. In the decoder, four convolutional blocks are cascaded to gradually upsample inherent features and fused them with those from the encoder. The entire structure is elaborated in the following. Furthermore, the range-Doppler matrix is originally a complex tensor with size $512 \times 256 \times 16$ in range, Doppler, and receiver antenna dimensions respectively. Herein, we split it into real and imaginary parts, reshaped, and formed it to shape $256 \times 512 \times 32$ as the input. The network predicts the free space segmentation of the surrounding area in the polar coordinate system as the output. The output size 256×512 covers $[-45^\circ, 45^\circ]$ azimuth and $[0, 100 \text{ m}]$ range, respectively.

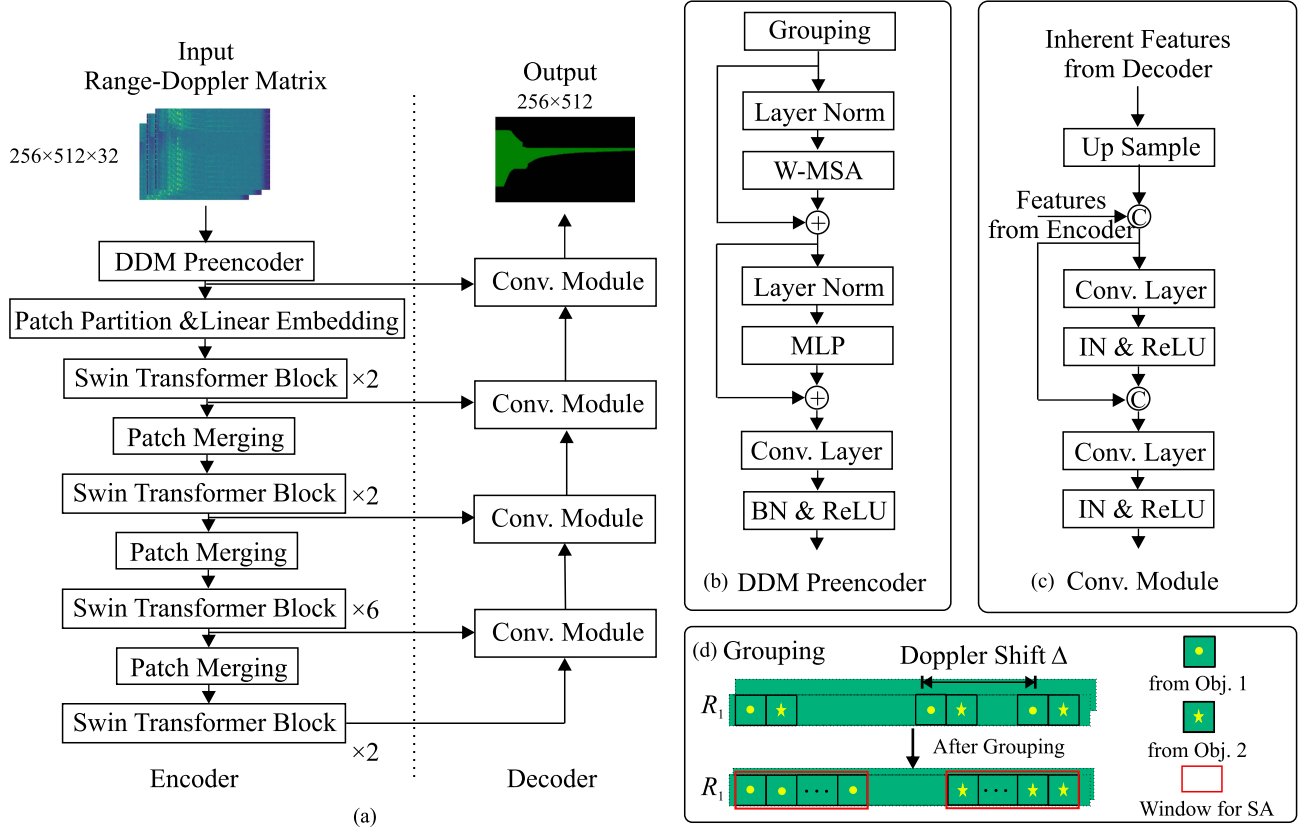


Fig. 3. Structure of the radar raw data-based, data-driven ISM network is depicted in (a). The network consists of an encoder, which contains DDM Preencoder in (b) and a Swin Transformer backbone, and a decoder, which is cascaded by four Conv. Modules, presented in (c). Subfigure (d) demonstrates the grouping operation in DDM Preencoder.

The first part of the encoder is a DDM Preencoder, depicted in Fig. 3(b). Although the Swin Transformer backbone applies self-attention solely within the local window scope to keep the computation linearly to the data size, a shift window design can facilitate the feature exchange among neighboring windows so that the network achieves a global understanding indirectly. However, due to the adoption of DDM technology, in the range-Doppler matrix, one received signal repeats 12 times, which is the number of transmitter antennas, along the Doppler axis, and the Doppler shift between two repeated, adjacent signal is 16, denoted Δ . Facing such distributed signals, directly introducing the Swin Transformer backbone to the range-Doppler matrix incurs worse results. Therefore, we followed the design ethos of the Swin transformer that applies self-attention within the local window, and proposed a DDM Preencoder. In this preencoder, we supplemented a grouping operation, specially aimed at DDM, illustrated in Fig. 3(d). In subfigure (d), assuming at range position R_1 , signals from objects 1 and 2 are interleaved and distributed among the Doppler axis before grouping. During this operation, signals distanced by one or multiple times of the known Doppler shift Δ are grouped together. Consequently, signals from the same object are clustered together and the following W-MSA is applied to each group denoted in the red window. The self-attention is formulated as

$$\text{Attention}(Q, K, V) = \text{SoftMax} \left(\frac{QK^T}{\sqrt{d}} + B \right) V \quad (5)$$

where Q, K, V are query, key, and value obtained from projections of signals in each window, and B denotes its relative position within a window. After W-MSA and the related layer normalization, multi-layer perceptron (MLP), a convolutional layer with 1×1 kernel size is utilized to adjust the channel dimension to C , which is equal to 48 in this work.

After the DDM Preencoder, a hierarchical Swin Transformer backbone further extracts prominent features from the data with different resolutions. Since the patch partition is configured with 4×4 patch size along with stride 2, the height and width of feature maps are down-sample to its half. Within each Swin Transformer Block, W-MSA and shift window-based multihead self-attention (SW-MSA) are alternately computed in feature maps. Furthermore, a patch merging operation condenses the height and width of features to half of their size and simultaneously doubles the depth. By means of the Swin Transformer backbone, hierarchical feature maps are generated, and those along with the output from the DDM Preencoder are delivered to the decoder.

Considering the requirement on computational efficiency and model size, the decoder is structured by four convolutional modules in series. In each module, its inherent features are first upsampled and then concatenated with data through a skip connection direct from the encoder. Afterwards, two convolutional layers with kernel size 3×3 along with instance normalization and ReLU operations process the data. Finally, the output from

the decoder is the predicted OGM with size 256×512 in the polar coordinate system. In addition, ablation experiments are conducted in Section V-D to analyze the proposed network, and a qualitative comparison with a fully convolutional CM model proposed from our previous work [38], is presented in Section V-E.

We conducted the training with pytorch in a workstation equipped with a Xeon W-1390P CPU, a Nvidia RTX 5000 GPU and 64 GB RAM. Furthermore, we selected the Adam optimizer [55], batch size 4, and a step learning rate schedule with initial rate 4×10^{-4} and a decay of 0.9 every 10 epochs. The training lasted altogether 60 epochs.

The loss function $\mathcal{L}_{\text{total}}$ is formulated in (6) and consists of the binary cross entropy and dice losses [56]. To address the classes imbalance and better predict road details, dice loss is adopted as the addition term. The weights for two losses α_{bce} and α_{dice} are 0.4 and 0.6, respectively. The binary cross entropy and dice losses are presented in (7) and (8), where y and $\hat{y}$ are the label and prediction within the data $\mathcal{G}$.

$$\mathcal{L}_{\text{total}} = \alpha_{\text{bce}} \cdot \mathcal{L}_{\text{bce}} + \alpha_{\text{dice}} \cdot \mathcal{L}_{\text{dice}} \quad (6)$$

$$\mathcal{L}_{\text{bce}} = - \sum_{\mathcal{G}} y \cdot \log(\hat{y}) \quad (7)$$

$$\mathcal{L}_{\text{dice}} = 1 - \frac{2 \cdot \sum_{\mathcal{G}} y \cdot \hat{y}}{\sum_{\mathcal{G}} y + \sum_{\mathcal{G}} \hat{y}} \quad (8)$$

The proposed model has about 12.4 M parameters in total and requires 198 GFLOPS for one inference. Temporally, it takes approximate 64 ms to predict one measurement. Considering that the working cycle of an automotive radar is around 60 ms, the model has the potential to be applied in real-time.

V. EXPERIMENTAL RESULTS

In Sections V-A and V-B, we comprehensively evaluated the free space segmentation performances of these three ISMs listed in Fig. 1 through the following metrics: accuracy, precision, recall, IoU, false alarm rate (FAR) and F_1 score. F_1 score is defined in (10), where N denotes the number of measurement, and TP_i stands for true positive, TN_i for true negative, FP_i for false positive, and FN_i for false negative, formulated in (9). For a better ISM, it should achieve a lower score in metric FAR, while higher scores in other metrics. The classic, hand-crafted ISM is based on the work in [9], and we also implemented Occupancy Net, according to [11].

$$\begin{aligned} TP_i &= (y_i = 1) \text{ and } (\hat{y}_i = 1) & TN_i &= (y_i = 0) \text{ and } (\hat{y}_i = 0) \\ FP_i &= (y_i = 0) \text{ and } (\hat{y}_i = 1) & FN_i &= (y_i = 1) \text{ and } (\hat{y}_i = 0) \end{aligned} \quad (9)$$

$$F_1 = \frac{1}{N} \sum_{i=1}^N \frac{2TP_i}{2TP_i + FP_i + FN_i} \quad (10)$$

Afterwards, in Section V-C, we quantitatively compared our model with other data-driven models concerning the segmentation performances, and in Section V-D, ablation experiments are conducted to analyze our model. Finally, qualitative analyses

of prediction from our model with those from other models are presented in Section V-E.

A. Comparisons of Three ISMs

We first evaluated all three ISMs with the six metrics, covering the entire range scope $[0, 100 \text{ m}]$ in Fig. 4. As the odometry data for the entire dataset are not provided, we only utilized the hand-crafted ISM to predict the OGM for two traces in the validation data. Albeit with partial measurement, the results can still reflect the tendency explicitly and support the derived conclusions. Furthermore, for the hand-crafted ISM, the value $p(\mathbf{m}_{u,v} | z_{1:t}, x_{1:t})$ for cell (u, v) represents the posterior likelihood of occupancy. Hence, $1 - p(\mathbf{m}_{u,v} | z_{1:t}, x_{1:t})$ denotes the corresponding likelihood of free space. To assess the segmentation comprehensively, the prediction threshold for free space as horizontal axes in Fig. 4(a)–(l) varies from 0.2 to 0.8 with a 0.05 step.

Analyzing Fig. 4, we can observe two significant phenomena. First, the hand-crafted ISM is obviously inferior to its data-driven counterparts. Although the recall of the hand-crafted ISM is higher when the threshold is approximately lower than 0.6 in Fig. 4(c) and (f), its precision is much worse than those of the other two ISMs in (b) and (e). Moreover, in Fig. 4(i) and (l), F_1 that considers the precision and recall simultaneously, indicates that two data-driven models are superior concerning the free space segmentation task. Second, focusing on the two data-driven ISMs, although our model outperforms Occupancy Net, the differences are not so significant as those from [15]. Concretely, comparing Fig. 4(f) and (l), raw data-based model yields around 2% improvement concerning recall and F_1 , and almost 4% upgrade regarding IoU in (j). However, both models are rather close considering accuracy and FAR.

B. Comparisons of Two Data-Driven ISMs

Based on the analyses in Section V-A, the improvement with the raw data is not significant as stated in [15]. Due to the observation that distance affects the segmentation performance of two data-driven ISMs to different extents, we further partitioned the space into $[0, 50 \text{ m}]$ and $[50 \text{ m}, 100 \text{ m}]$ to evaluate the segmentation results. The outcomes are presented in Fig. 5.

Analyzing the results for $[0, 50 \text{ m}]$ and $[50 \text{ m}, 100 \text{ m}]$ in Fig. 5, we yielded inconsistent findings with [15].

For one thing, generally, two data-driven ISMs are on par in the scope $[0, 50 \text{ m}]$. Although our approach overtakes Occupancy Net in five metrics except recall, the improvement is slight. The reasons could be twofold. First, the input data for the raw data-based ISM are range-Doppler matrices, and the model must learn the latent angular information from the channel dimension. Within the $[0, 50 \text{ m}]$ scope, compared with traditional methods such as FFT or MUSIC, this learning-based method does not show obvious advantages concerning accuracy. In addition, the detection generated by the CFAR can still accurately reflect the surrounding environment within this closer scope. For another, our raw data-based ISM overtakes the detection-based counterpart by a large margin in the $[50 \text{ m}, 100 \text{ m}]$ range for most of the

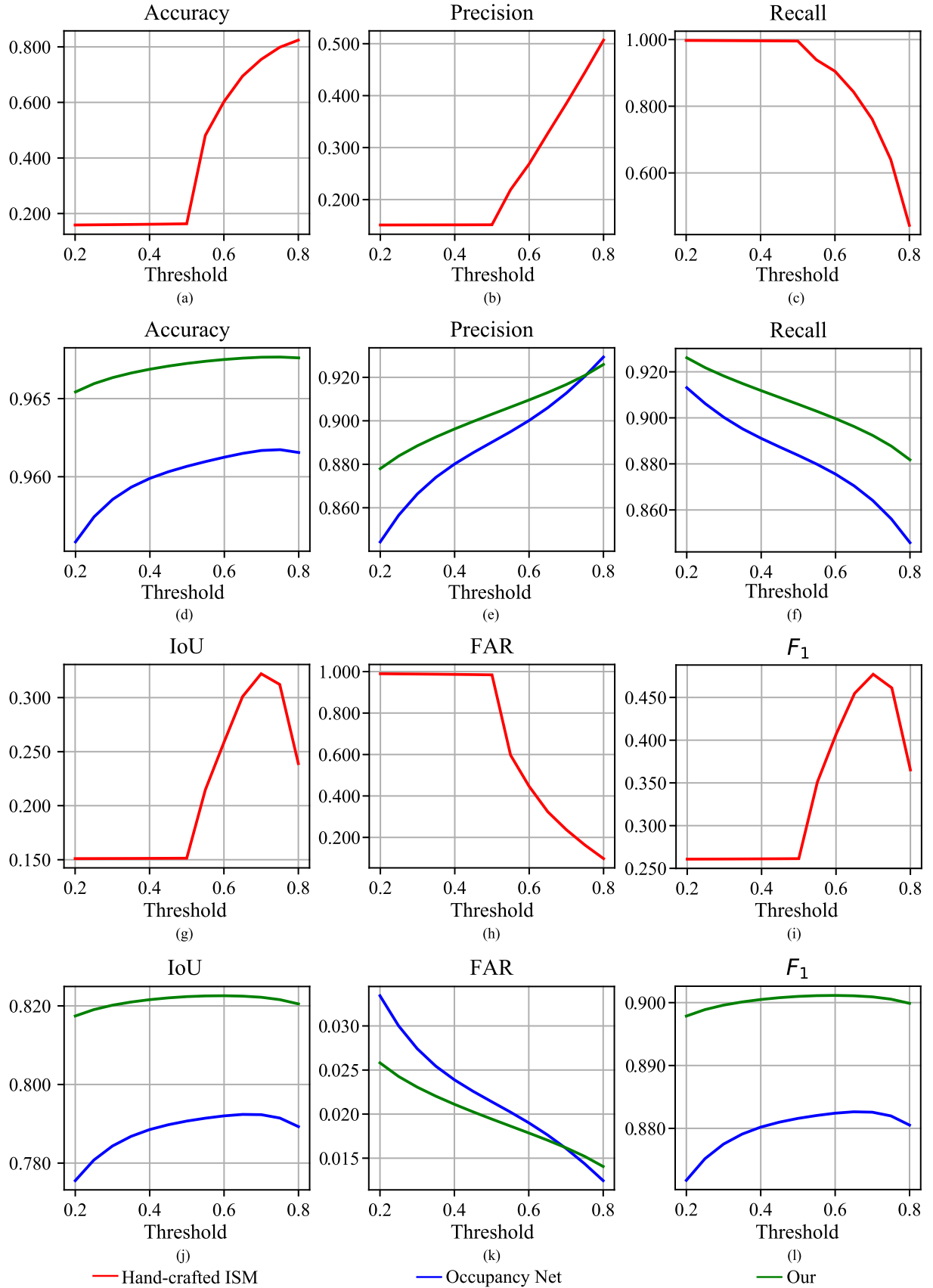


Fig. 4. Three ISMs evaluation with six metrics covering the entire range scope [0, 100 m]. Two data-driven ISMs are notably superior to their hand-crafted counterpart. Although our raw data-based ISM outperforms Occupancy Net, the difference are not as significant as those from [15].

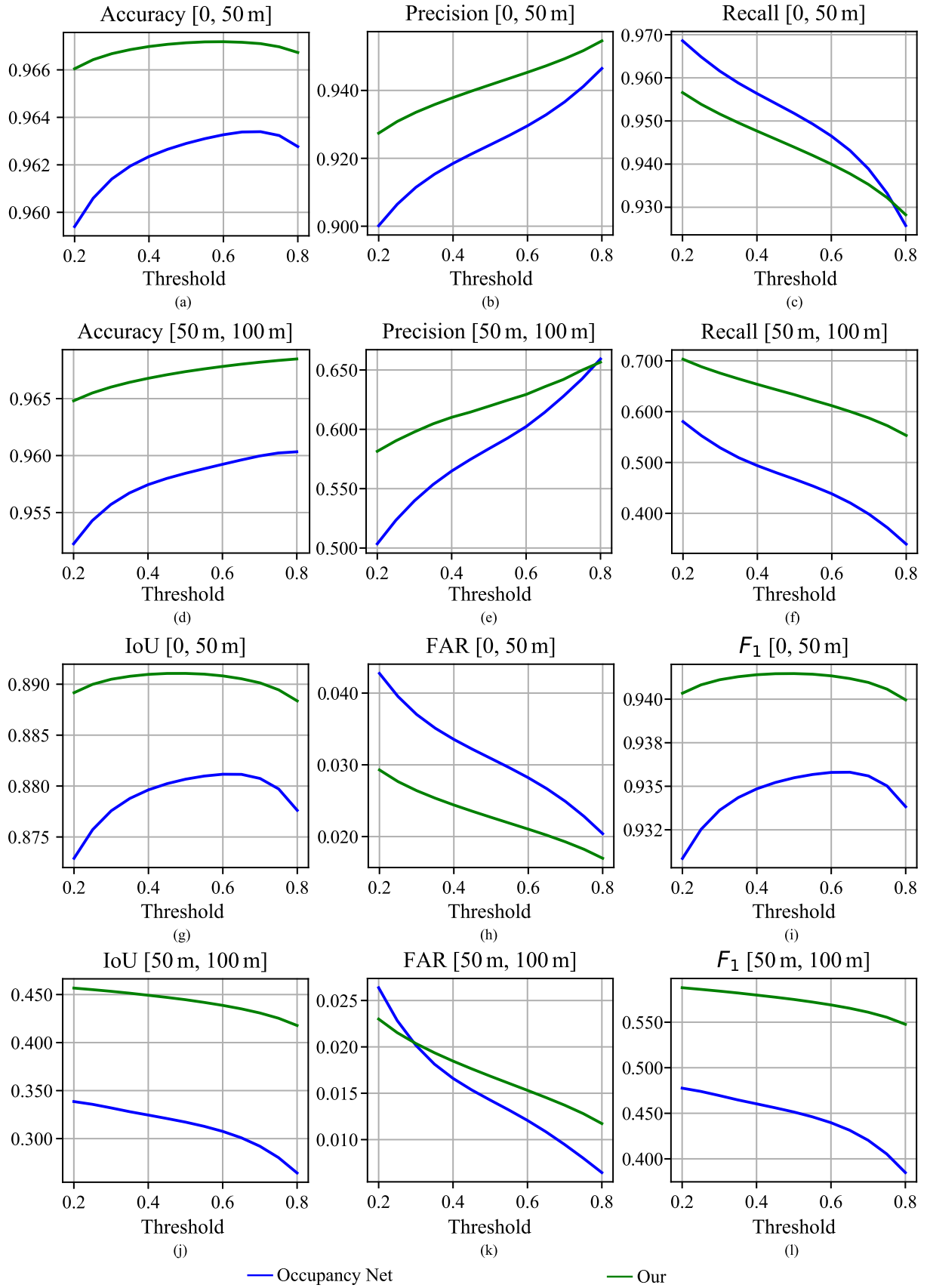


Fig. 5. Evaluation of two data-driven ISMs with six metrics in [0, 50 m] and [50 m, 100 m], respectively. Generally, two models achieve fairly close performance in [0, 50 m]. However, our raw data-based ISM dominates obviously in [50 m, 100 m], compared with Occupancy Net.

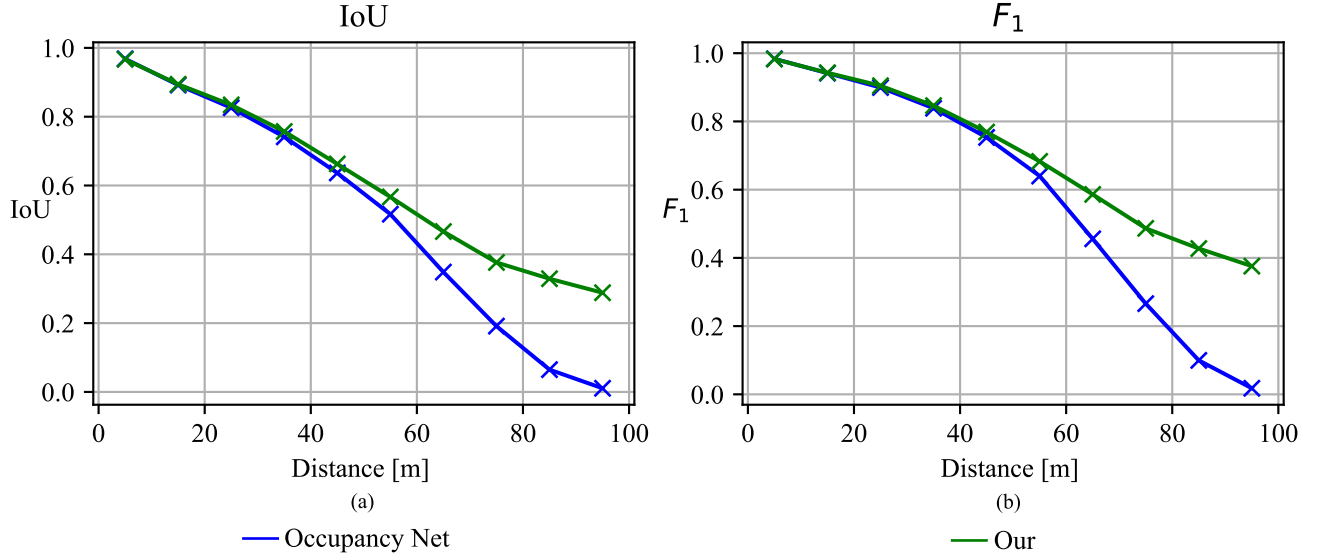


Fig. 6. Comparisons of two data-driven ISMs given fixed threshold concerning IoU and F_1 in (a) and (b), respectively. Obviously, within 50 m, two models are on par with each other. However, in the range [50 m, 100 m], our model overtakes Occupancy Net by a large margin.

metrics. Especially concerning IoU and F_1 in Fig. 5(j) and (l), our model exceeds Occupancy Net by about **0.1**.

To further confirm these findings, we investigated the performance variation of both ISMs with respect to the distance. Herein, we first fixed the threshold as 0.6 and partitioned $\mathcal{G}$ along the range direction into ten subareas equally. Instead of calculating all six metrics, we selected the most informative IoU and F_1 for evaluation. Furthermore, the calculated metrics values are assigned to the middle range of each subarea, and the results are presented in Fig. 6. In Fig. 6(a), the IoUs of both ISMs decrease when increasing the distance. Within the 50 m range, both ISMs are on the same level, while more than 50 m raw data-based ISM prevails notably and finally achieves about **0.3** higher than the counterpart. The observation of F_1 in Fig. 6(b) is totally consistent with that of IoU, and the differences are even enlarged to about **0.4** at the end.

Considering the results from Figs. 5 and 6, we find that in near range ($[0, 50\text{ m}]$), both data-driven ISMs are almost on par with each other, while in the far range ($[50\text{ m}, 100\text{ m}]$), our raw data-based one defeats the detection-based Occupancy Net significantly.

C. Segmentation Performance Comparison Among Data-Driven ISMs

In Table II, we compared the segmentation performance of several data-driven ISMs with ours, quantitatively. Herein, T-FFTRadNet [57] is also a Swin Transformer-based segmentation network. CM model was proposed in our previous work [38], and its detection branch is deleted to adjust to the segmentation task. Our model transcends all other models obviously in not only IoU but also F_1 metrics.

D. Ablation Analysis

To analyze the effect of the DDM Preencoder and Swin Transformer Backbone in our proposed network, we conducted

TABLE II
SEGMENTATION PERFORMANCE COMPARISON

Model	Input	IoU(%) $\uparrow$	F_1 (%) $\uparrow$
Occupancy Net [11]	Det.	78.0	87.7
T-FFTRadNet [57]	RD	79.1	88.2
CM [38]	RD	80.0	88.9
Our	RD	82.2	90.4

In Input column, Det. and RD mean detection and range-Doppler matrix, respectively. The best results are in bold.

TABLE III
ABLATION ANALYSIS

DDM Preencoder	MIMO Preencoder [17]	Swin Transformer Backbone	IoU(%)	F_1 (%)
-	-	$\checkmark$	79.3	87.1
$\checkmark$	-	$\checkmark$	82.2	90.4
-	$\checkmark$	$\checkmark$	80.7	89.5
$\checkmark$	-	-	79.4	88.6

ablation experiments in Table III. Herein, MIMO Preencoder, proposed by [17], is based on dilated convolution [58], and it sets Doppler shift Δ as the dilation rate. For the last experiment, the Swin Transformer Backbone was replaced by a convolutional encoder from [38].

Analyzing the first three experiments, we can find that the proposed preencoder is more effective than dilated convolution-based counterpart in extracting distinct features from the range-Doppler matrix, which is based on DDM technology. In addition, the first two and last experiments show the obvious improvement of DDM Preencoder and the Swin Transformer backbone to our model.

E. Qualitative Analysis

For qualitative analysis, Fig. 7 demonstrates three examples (I–III) in city, highway, and country road scenarios respectively to compare the performances of two data-driven ISMs. Each example contains the segmented camera image, range-Doppler

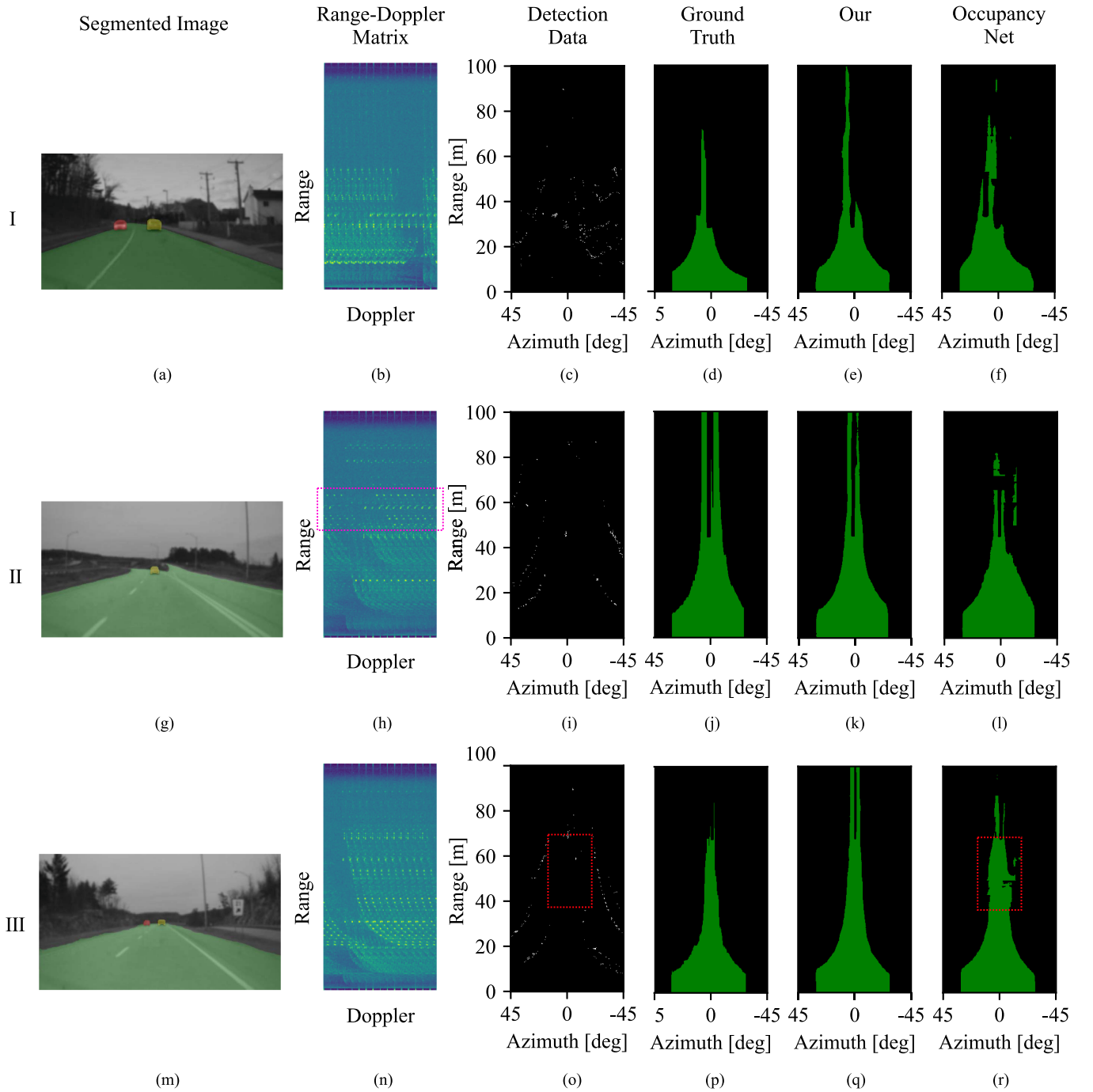


Fig. 7. Three examples (I–III) recorded in city, highway, and country road scenarios respectively, to compare the free space segmentation performances of two data-driven ISMs. Each example contains the segmented camera image, range-Doppler matrix, detection data in the polar coordinate system, OGM ground truth, the prediction of our raw data-based ISM and Occupancy Net, alphabetically. In all examples, our raw data-based ISM is more precise, especially when the range is beyond 50 m. In addition, it is more robust against the noise signal and scenarios variation. More specifically, our model is immune to the noise signal marked in (h) with a dash line rectangle, while the Occupancy Net fails at this area in (l). Regarding the scenario variation, in example III, the road boundary of the country road generates extremely sparse detections in the dash line labeled area in (o). This leads to the false prediction in (r), while our ISM still works normally.

matrix, detection data, OGM ground truth, and the predictions from two ISMs, alphabetically.

Analyzing these examples, we can observe two strengths of our ISM. First, it performs more robustly against scenario variations and interference such as noise or ground reflection. For instance, the signals marked by the dash line in Fig. 7(h) are rather noisy. Our raw data-based ISM is immune to the

interference and predicts correctly, while Occupancy Net fails in this area. In addition, in example III, the free space is delimited by the grasses and plants instead of the guard rails. Consequently, detection within the dash line rectangle in Fig. 7(o) is extremely sparse, and it leads to the false prediction in Fig. 7(r). In contrast, in Fig. 7(q), raw data-based method works stably under this challenging scenario. Furthermore, raw data-based ISM is more

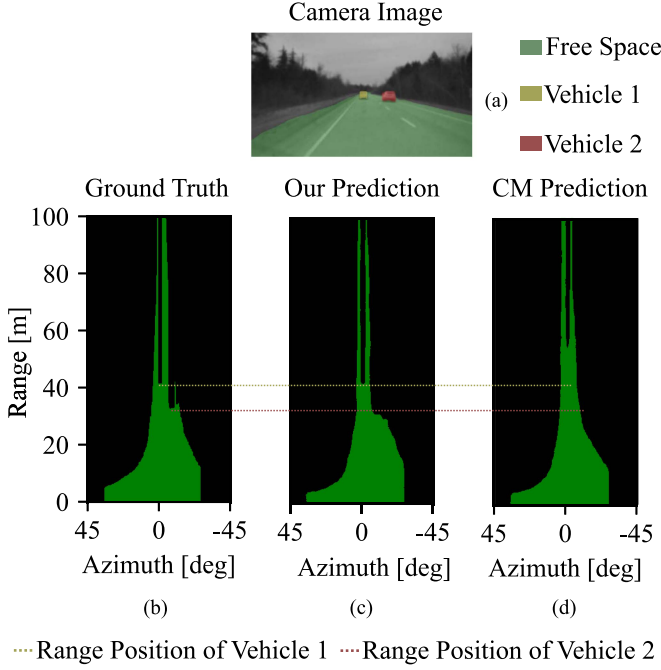


Fig. 8. Comparison of two raw data-based ISMs. Our model achieved a more precise prediction in not only range but also angular directions. Concretely, the prediction of our model for vehicle 1 in range along with that for vehicle 2 in angular direction, are closer to ground truth than those of the CM model.

reliable and more precise at a farther range. More specifically, its predictions are closer to the ground truth in all examples, when the range is above 50 m.

Furthermore, in Fig. 8, we compared the prediction result from our model with that of a fully convolutional CM model. In this instance, our model achieves higher range accuracy because its occupancy prediction for vehicle 1 is almost the same as the ground truth. In contrast, the prediction from CM model shows an obvious deviation in range direction. In addition, when observing vehicle 2, although the predictions of range from both models are similar, our model presents a more precise angular estimation capability.

VI. DISCUSSION

A. Radar-Based Occupancy Flow

In the AD scenarios, it is definitely necessary to distinguish between static and moving objects since these two kinds of objects should be handled differently. For classic hand-crafted ISM, the detection from dynamic objects is abandoned, due to the discrepancy in the accumulation. Because of a static world assumption, the Occupancy Net does not address this issue either. In principle, our raw data-based, data-driven ISM is based on a single measurement and does not differentiate between static and dynamic objects. However, this issue can be solved given a sequence of OGMs, and we call it radar-based occupancy flow.

To facilitate understanding, we utilized solely two predicted OGMs, symbolized as $\hat{\mathcal{G}}_{t_0}^c$ and $\hat{\mathcal{G}}_{t_0+\Delta t}^c$, from two adjacent radar data at moment t_0 and $t_0 + \Delta t$, where the hat denotes prediction,

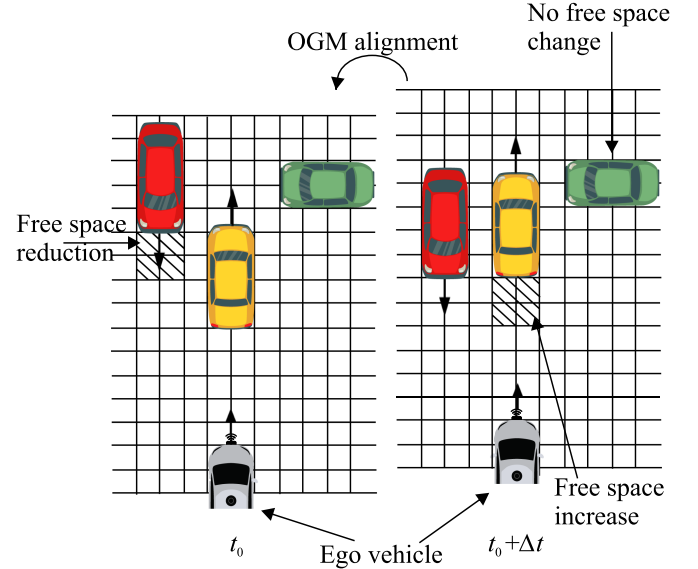


Fig. 9. Radar-based occupancy flow explanation. Comparing $\hat{\mathcal{G}}_{t_0}^c$ and $\mathcal{T}_{t_0+\Delta t}^{t_0}(\hat{\mathcal{G}}_{t_0+\Delta t}^c)$, we can observe the reduced free space due to the oncoming red vehicle. Similarly, the departing yellow vehicle causes a increase of free space. For the parked green vehicle, its surrounding free space does not change within Δt the interval.

and Δt is the measurement interval. In addition, the movement of ego vehicle within Δt is assumed to be known. Usually, the inertial measurement unit (IMU) of the vehicle can directly provide this odometry information. Nevertheless, the ego motion can also be estimated by using the data from radar [59], [60] or lidar [61]. Since it is missing in the RADIAL dataset, we employed the iterative closet point algorithm [62] to synchronized lidar data to estimate the movement of the ego vehicle within Δt . Based on this, we could define the transformation of the vehicle coordinate system from moment $t_0 + \Delta t$ to t_0 as $\mathcal{T}_{t_0+\Delta t}^{t_0}$.

Generally, the radar-based occupancy flow works in three steps. First, to promote the following processing, the predicted OGM must be converted from the polar coordinate system to the Cartesian, denoted as $\hat{\mathcal{G}}_{t_0}^c$ and $\hat{\mathcal{G}}_{t_0+\Delta t}^c$. After that, relying on $\mathcal{T}_{t_0+\Delta t}^{t_0}, \hat{\mathcal{G}}_{t_0+\Delta t}^c$ can be aligned with $\hat{\mathcal{G}}_{t_0}^c$ in the ego vehicle coordinate system of moment t_0 , represented by $\mathcal{T}_{t_0+\Delta t}^{t_0}(\hat{\mathcal{G}}_{t_0+\Delta t}^c)$. In final step, the motion information of objects in the OGM can be derived through a comparison of $\hat{\mathcal{G}}_{t_0}^c$ and $\mathcal{T}_{t_0+\Delta t}^{t_0}(\hat{\mathcal{G}}_{t_0+\Delta t}^c)$. For instance, Fig. 9 demonstrates the situations for the oncoming, departing, and parking vehicles. For the red oncoming vehicle, since it is approaching the ego vehicle, the free space reduction can be observed when comparing $\hat{\mathcal{G}}_{t_0}^c$ and $\mathcal{T}_{t_0+\Delta t}^{t_0}(\hat{\mathcal{G}}_{t_0+\Delta t}^c)$. Analogously, the departing yellow vehicle generates extra free space, and for the parked green one, the free space should be consistent. Practically, the comparison can be realized by a reduction operation.

Fig. 10 presents the results of occupancy flow to one measurement. Before reduction, in $\hat{\mathcal{G}}_{t_0}^c$ and $\mathcal{T}_{t_0+\Delta t}^{t_0}(\hat{\mathcal{G}}_{t_0+\Delta t}^c)$, 1 denotes free, while 0 means the location is occupied. After reduction, the extra free space marked by the red dash line can be clearly recognized in Fig. 10(b). In this regard, it is consistent with

TABLE IV
ISMS COMPARISONS

	Type	Input	No. frame(s)	Independent assumption	Moving object	Robustness against		Free space segmentation performance	Computation
						Doppler ambiguity	noise & clutter		
Classic	hand-crafted	detection	multiple	yes	no	bad	bad	bad	low
Occupancy Net	data-driven	detection	single	no	static world assumption	excellent	good	good	medium
Our	data-driven	range-Doppler matrix	single	no	occupancy flow	excellent	excellent	excellent	high

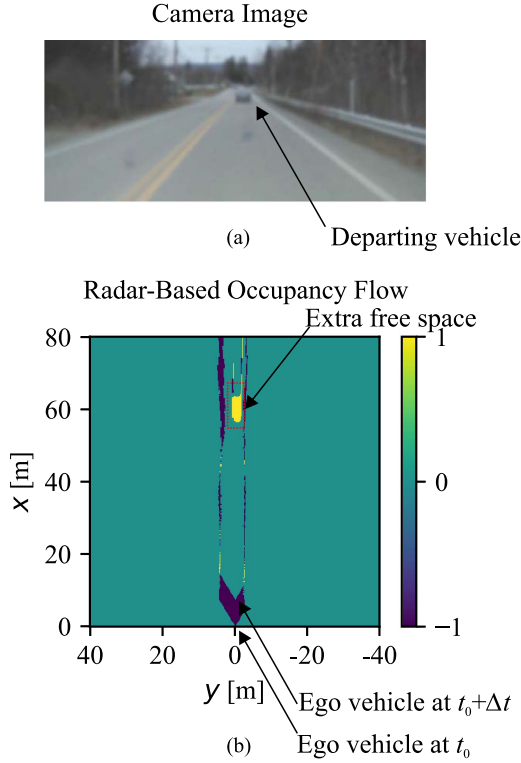


Fig. 10. Radar-based occupancy flow example. (a) shows the synchronized camera image. Originally, in $\hat{\mathcal{G}}_{t_0+\Delta t}^c$ and $\hat{\mathcal{G}}_{t_0}^c$, 1 denotes *free*, and other areas are assigned with 0. (b) is the result of reduction between $\hat{\mathcal{G}}_{t_0}^c$ and $\mathcal{T}_{t_0+\Delta t}^{t_0}(\hat{\mathcal{G}}_{t_0+\Delta t}^c)$. In (b), the extra free space induced by the departing vehicle that can be confirmed in (a), is marked by the red dash line. At two sides of the marked area, there are two thin stripes with a non-zero value. The strip is relatively minor compared with the aforementioned extra free space, and this discrepancy can be contributed to certain factors, such as imprecise alignment of $\hat{\mathcal{G}}_{t_0+\Delta t}^c$ and $\hat{\mathcal{G}}_{t_0}^c$ or OGM prediction error. In addition, the movement of the ego vehicle accounts for the area with value -1 at the origin.

the departing vehicle observed in the camera image in (a). In addition, the area close to the origin with the value -1 is caused by the motion of the ego vehicle. At two sides of the extra free space caused by the departing vehicle, there are two thin stripes with nonzero values. The discrepancy can arise from several reasons, such as imprecise alignment of $\hat{\mathcal{G}}_{t_0+\Delta t}^c$ and $\hat{\mathcal{G}}_{t_0}^c$, and false prediction in OGM. However, compared with the extra free space caused by the departing vehicle, this discrepancy is still negligible. In addition, the majority area with value 0 in (b) means that no change happens in the interval Δt .

B. Discussion of Three ISMs

In Table IV, three ISMs are extensively compared. Different from the other two, the hand-crafted ISM assumes that the

likelihood of occupancy for each cell is independent and accumulates solely static detection from different moments. Since it abandons the moving detection, it is rather sensitive to the issue like Doppler ambiguity. For instance, the authors in [38] illustrates the severe result when the ego vehicle drives faster than the velocity unambiguous scope. Furthermore, due to its simplicity, the hand-crafted ISM suffers from noise and clutter and has the worst performance for free space segmentation.

The two data-driven ISMs, utilizing detection and the range-Doppler matrix of single measurement as the inputs respectively, learn the underlying joint probability function from the data, and thus the independent assumption for OGM is no longer held. Although both assume a static world, the raw data-based ISM can further utilize the occupancy flow to determine the moving object. Owing to the learning process, both models are more robust against the interference, such as Doppler ambiguity, noise, and clutter, and the performances for free space segmentation are superior to the hand-crafted counterpart. The raw data-based ISM achieves the best performances in all aspects, except the computational load. Note that the computation merely refers to the prediction of OGM. Since the raw data-based ISM employs a range-Doppler matrix instead of detection as input, it can actually save intensive computation for the angle estimation.

Based on the results in Section V and Table IV, we can derive the following findings. First, the data-driven ISM is more capable than the hand-crafted counterpart and thus having far superior performance in the road segmentation task. Furthermore, our model outperforms Occupancy Net, and the advantage in the long range is significantly obvious. Considering that high levels autonomous driving, such as L3 and above, impose higher requirements on perception systems, advantages of raw data-based ISM in the long range are of great significance. In addition, despite the inferiority of detection-based Occupancy Net, it is on par with our model within the near range. Its failure in the far range can be mainly contributed to the extremely sparse detection therein. This infers that the methods to improve the radar detection quality, such as [63], [64], are in a promising direction to further improve the performance of this kind of ISM.

VII. CONCLUSION

In this work, we have proposed a novel radar raw data-based ISM that has the potential to be applied in real time to achieve the road segmentation task. To the best of our knowledge, we are the first to conduct a systematic assessment and comparison among these three ISMs. With our results and analyses, the questions of whether to employ raw data-based ISM and when have been well addressed. For the radar of RADIAL dataset, raw data-based, data-driven ISM outperforms the detection-based

counterpart significantly in the distant range ([50 m, 100 m]), while in the near range (< 50 m) both are on par. In addition, the radar-based occupancy flow is designed to be able to recognize moving objects from the simultaneous OGM. As future work, an extension to the 3D OGM with radar raw data can be a promising direction.

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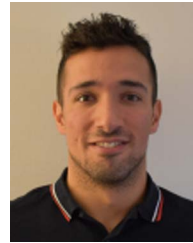
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Yi Jin (Graduate Student Member, IEEE) was born in Kunshan, China, in 1992. He received the M.Sc. degree in electromobility from the University of Stuttgart, Stuttgart, Germany, in 2019. He is currently working toward the Ph.D. degree with the Institute of Microwaves and Photonics (LHFT), Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU), Erlangen, Germany. In 2019, he joined the Institute of Microwaves and Photonics, Friedrich-Alexander-Universität Erlangen-Nürnberg. His research interests include (self-)supervised learning with automotive radar and sensor fusion.



Marcel Hoffmann (Graduate Student Member, IEEE) was born in Witten, Germany, in 1994. He received the B.Sc. and M.Sc. degrees in electrical engineering in 2016 and 2018, respectively, from the Friedrich-Alexander University Erlangen-Nürnberg, Erlangen, Germany, where he is currently working toward the Ph.D. degree with the Institute of Microwaves and Photonics (LHFT). In 2018, he joined the LHFT, Friedrich-Alexander University Erlangen-Nürnberg. His research interests include radar signal processing and automotive radar applications, including new SAR and SLAM approaches.



2019, he has been with BMW Group as a Radar Expert in autonomous driving and advanced driver assistance projects. His research interests include signal processing algorithms, automotive radar performance, sparse array design, convex optimization, beamformer development and game theoretic methods, within the radar network framework and wireless communications.



Juan-Carlos Fuentes-Michel received the Ph.D. degree from the Clausthal University of Technology, Clausthal-Zellerfeld, Germany, in 2008. He is currently a Radar Technology Expert for driver assistance systems and autonomous driving with the BMW Group and the author of several publications on wireless positioning and radar signal processing. During his career, he has contributed to the industrialization of different radar components in the automotive industry, from the initial concept phase till mass implementation.



Martin Vossiek (Fellow, IEEE) received the Ph.D. degree from Ruhr-Universität Bochum, Bochum, Germany, in 1996. In 1996, he joined Siemens Corporate Technology, Munich, Germany, where he was the Head of the Microwave Systems Group, from 2000 to 2003. Since 2003, he has been a Full Professor with Clausthal University, Clausthal-Zellerfeld, Germany. Since 2011, he has been the Chair of the Institute of Microwaves and Photonics (LHFT), Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU), Erlangen, Germany. He has authored or coauthored more than 350 articles. His research has led to more than 100 granted patents. His research interests include radar, microwave systems, wave-based imaging, transponder, RF identification, communication, and wireless locating systems. He is a member of the German National Academy of Science and Engineering (acatech) and of the German research foundation (DFG) review board. He is a member of the IEEE Microwave Theory and Technology (MTT) Technical Committees MTT-24 Microwave/mm-wave Radar, Sensing, and Array Systems, MTT-27 Connected and Autonomous Systems (as founding chair), and MTT-29 Microwave Aerospace Systems. He is also serving on the advisory board of the IEEE CRFID Technical Committee on Motion Capture & Localization. He was the recipient of numerous best paper prizes and other awards. In 2019, he was awarded with the Microwave Application Award from the IEEE MTT Society (MTT-S) for Pioneering Research in Wireless Local Positioning Systems. Dr. Vossiek has been a Member of organizing committees and technical program committees for many international conferences and he has served on the Review Boards for numerous technical journals. From 2013 to 2019, he was an Associate Editor for the IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES. Since October 2022, he has been an Associate Editor-in-Chief for IEEE TRANSACTIONS ON RADAR SYSTEMS.